

1. Aim and data

The aim of this project was to analyse 16S rRNA amplicon sequencing data from sample SRR13569534, targeting the V3–V4 variable regions and generated as Illumina paired-end reads and to identify which microorganisms are present in the sample and to present their relative taxon abundances. To achieve this and enable a direct comparison of methods, taxonomic profiling was performed using two approaches: DADA2 (denoising-based inference of ASVs followed by taxonomic assignment) and Kraken2 (k-mer-based read classification, refined for abundance estimation).

This made it possible to compare the extent to which the results of both tools are consistent at the level of higher taxonomic ranks and what differences appear with more detailed identification.

2. Quality profile before trimming

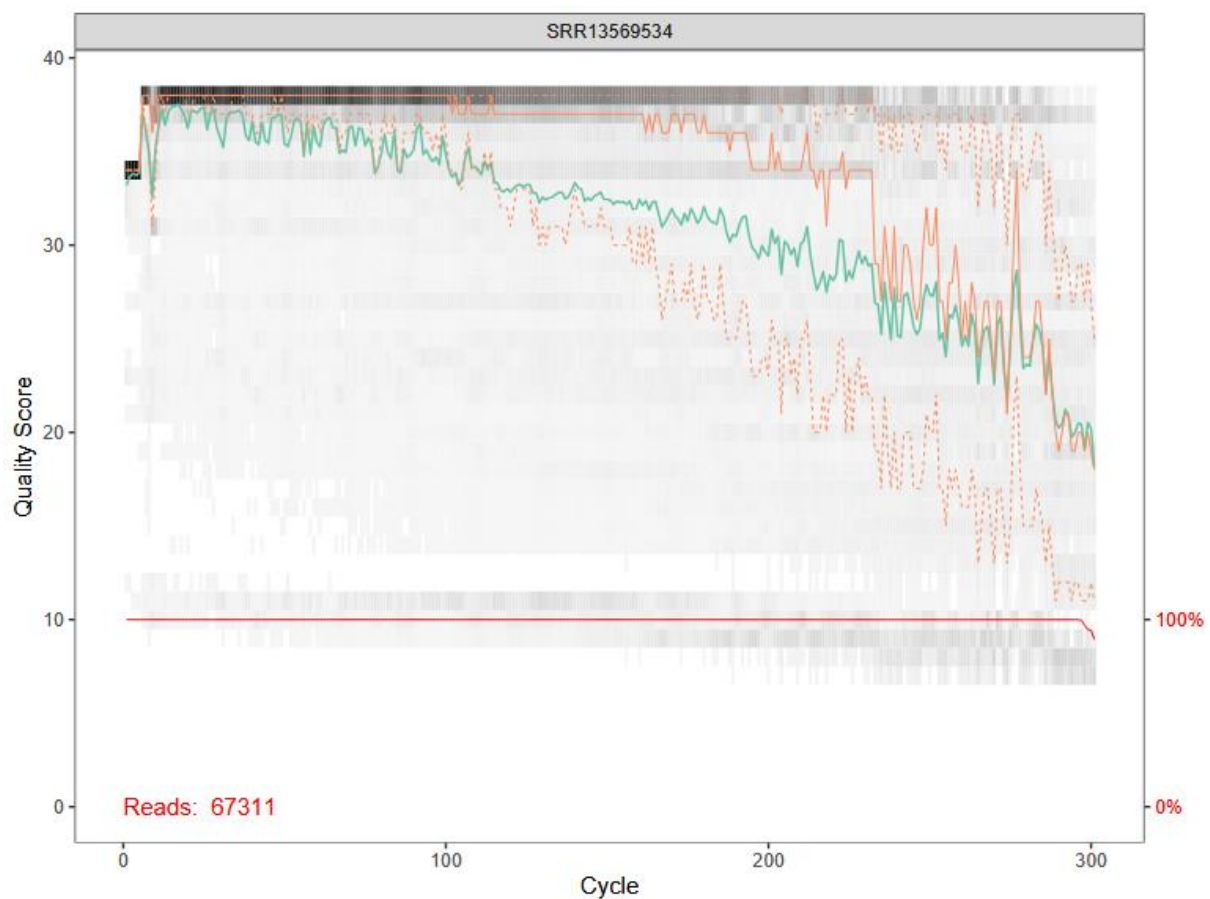


Fig. 1.1 Quality profile before trimming – forward

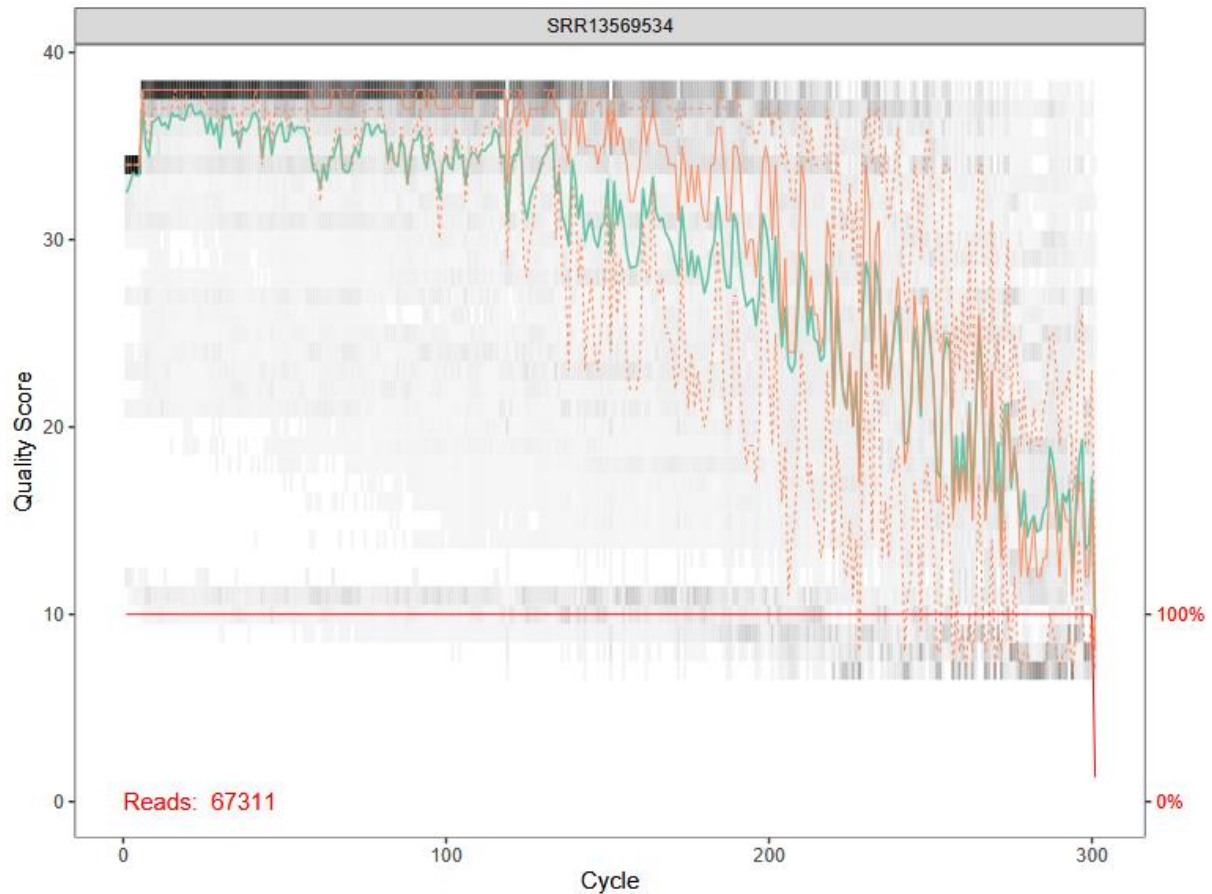


Fig. 1.2 Quality profile before trimming - reverse

In the quality profile plots for SRR13569534 before trimming, the forward reads maintain high base quality across most of the read length. For roughly the first 200-250 cycles the quality is stable, typically around Q30-Q38, and a more noticeable decline appears only toward the end, where the scores become lower and more variable. In contrast, the reverse reads deteriorate earlier: after about 180-220 cycles the quality drops more strongly and the spread increases, and the final part of the reads is clearly problematic.

This pattern indicated that the reverse reads required more aggressive trimming than the forward reads in order to remove the low-quality tail that would otherwise introduce errors into next DADA2 steps.

3. Trimming parameters

Based on the sequence quality profiles, separate trimming and filtering settings were applied for forward and reverse reads in the dada2: filterAndTrim step. The forward reads were truncated to 280 bp and the reverse reads to 220 bp, because these cutoffs remove the region where read quality begins to noticeably degrade while keeping the highest-quality part of each read. At the same time, the chosen truncation lengths were kept long enough to preserve sufficient overlap between paired reads for successful merging in later steps.

Filtering was primarily controlled using expected errors, with $\text{maxEE} = 2$ for forward reads (stricter, reflecting higher overall quality of forward) and $\text{maxEE} = 3$ for reverse reads (slightly more permissive, appropriate for the typically lower quality of reverse). Reads containing ambiguous bases were removed ($\text{maxN} = 0$), and truncation at very low quality was enabled ($\text{truncQ} = 2$) to cut reads at the first position with extremely low quality scores. No fixed trimming from the start or end was applied ($\text{trimLeft} = 0$, $\text{trimRight} = 0$). In addition, a minimum read length of 200 bp was enforced to remove unusually short reads that are more likely to cause downstream issues. After filtering and trimming, 34,553 out of 67,311 read pairs passed the step (approximately 51%), indicating a substantial removal of low-quality data while retaining a usable number of reads for next analysis.

4. Quality profile after trimming

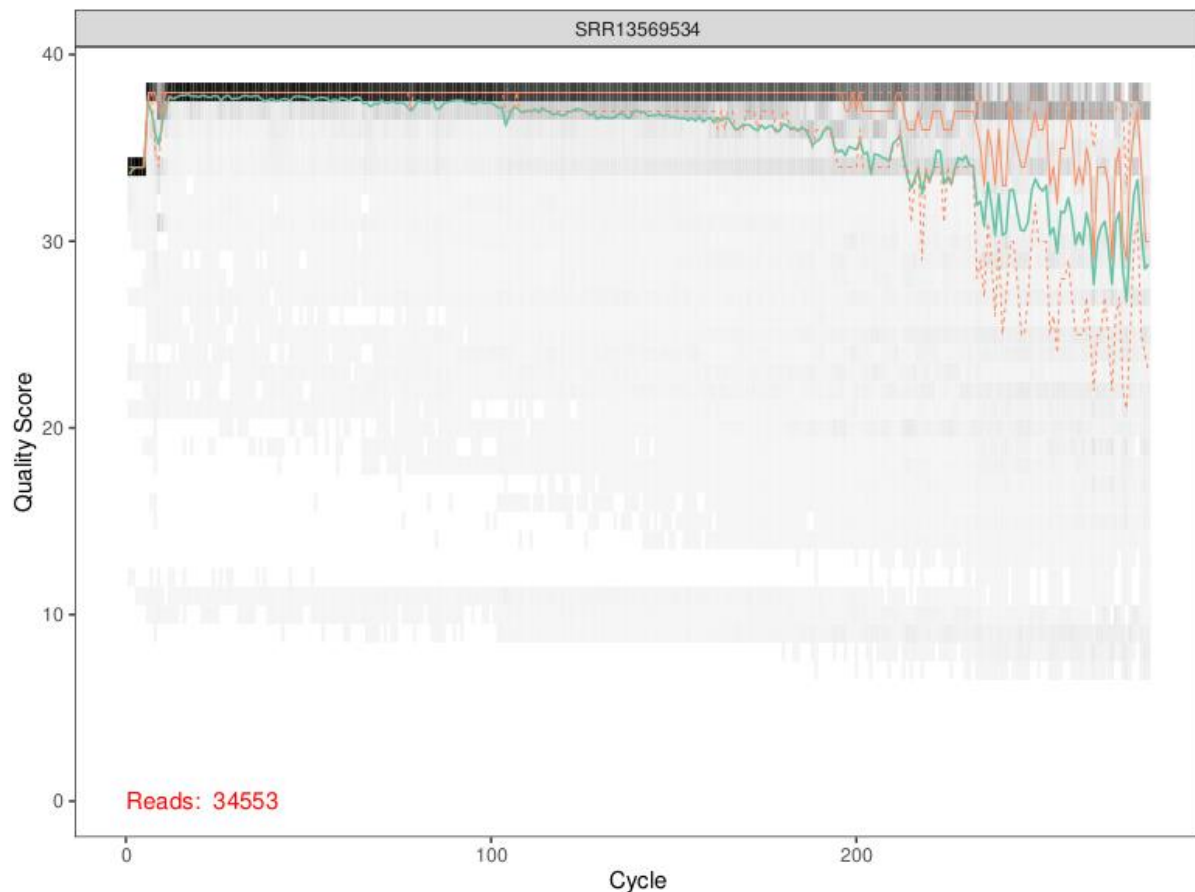


Fig. 2.1 Quality profile before trimming – forward

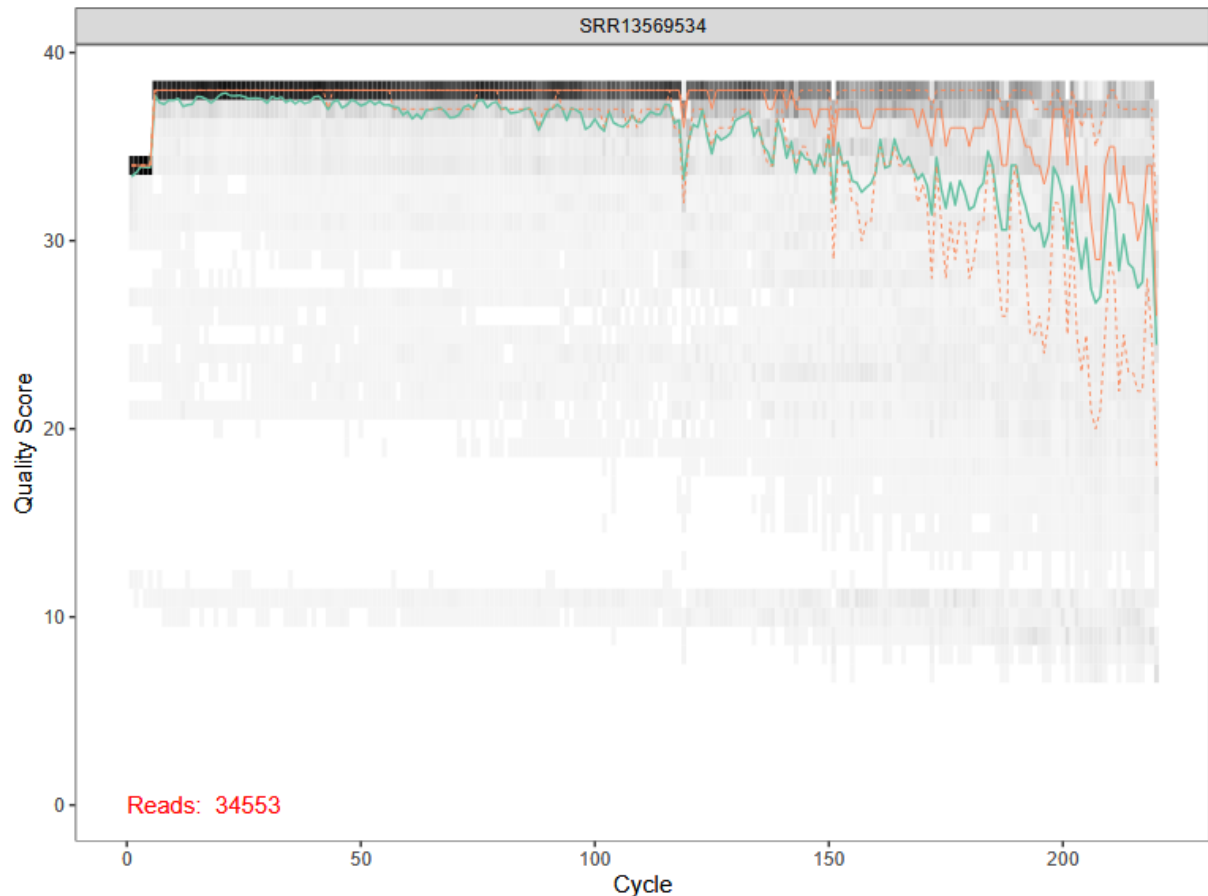


Fig. 2.2 Quality profile before trimming – reverse

After applying the selected filterAndTrim settings, the quality profiles show a clear improvement compared to the raw reads. For the forward reads (Fig. 1), the distribution is now dominated by consistently high scores across almost the entire retained length. The low-quality tail visible in the untrimmed data is effectively removed, and the median quality remains stable, with only a mild decrease close to the truncation point.

For the reverse reads, the improvement (Fig. 2) is even more noticeable, because they originally degraded earlier and more strongly. After trimming, the reverse profile keeps a high median quality across most cycles, and the strong drop and high variability that typically appears in the later part of reverse reads is minimized by cutting the read before that region.

MergePairs statistics confirmed that the chosen truncation lengths provided sufficient overlap for paired-end merging. The merged reads typically showed an overlap of several dozen base pairs (nmatch), while mismatches and indels in the overlap were negligible (nmismatch = 0, nindel = 0), indicating that low-quality read tails were successfully removed without compromising merge performance.

Overall, the post-trimming profiles show a tighter quality distribution and fewer very low-quality positions, which supports cleaner downstream denoising and more successful merging.

5. Dada2 vs Kraken2 - the analysis to taxon identification and visualizations of taxon abundancies

DADA2 and Kraken2 differ mainly in what they treat as the basic unit of analysis and how they assign taxonomy. DADA2 is an amplicon-focused pipeline: it denoises 16S reads to infer exact amplicon sequence variants (ASVs), removes sequencing errors and chimeras, and then assigns taxonomy to these ASVs using a reference database. This makes DADA2 well suited for high-resolution 16S community profiling, with results expressed as ASV abundances.

Kraken2 is a k-mer based classifier: it assigns taxonomy by matching short k-mer fragments from reads to a prebuilt reference database and classifying each read to the lowest common ancestor supported by those matches. With Bracken, these read-level classifications can be re-estimated into more robust abundance profiles at a chosen taxonomic rank (e.g., genus or species). In practice, DADA2 emphasizes error-corrected, ASV-level inference for amplicons, while Kraken2 emphasizes fast read classification based on k-mers and database content, which can lead to different taxonomic resolution and abundance estimates even on the same 16S V3–V4 dataset.

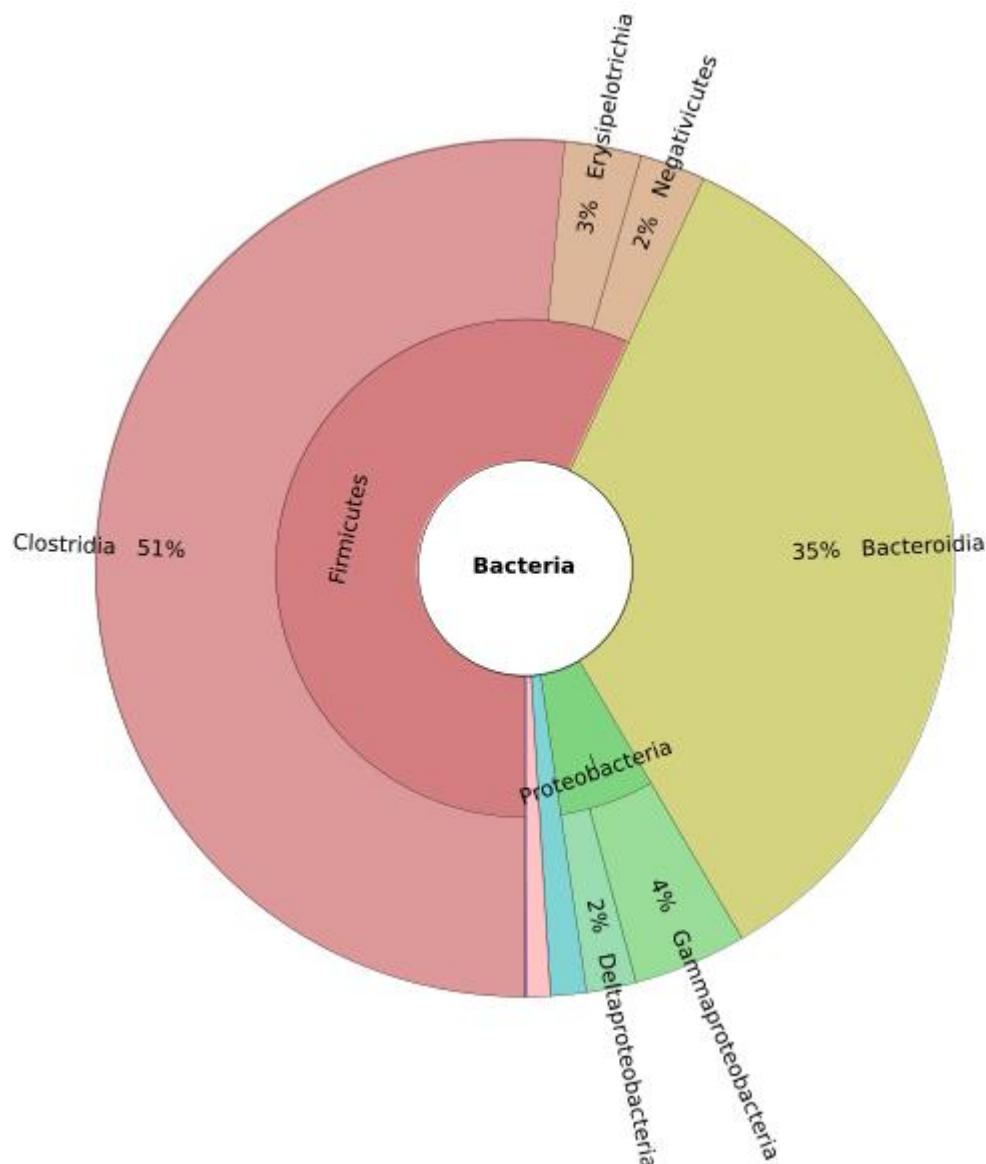


Fig 3.1 DADA2 – taxon relative abundance (Class level)

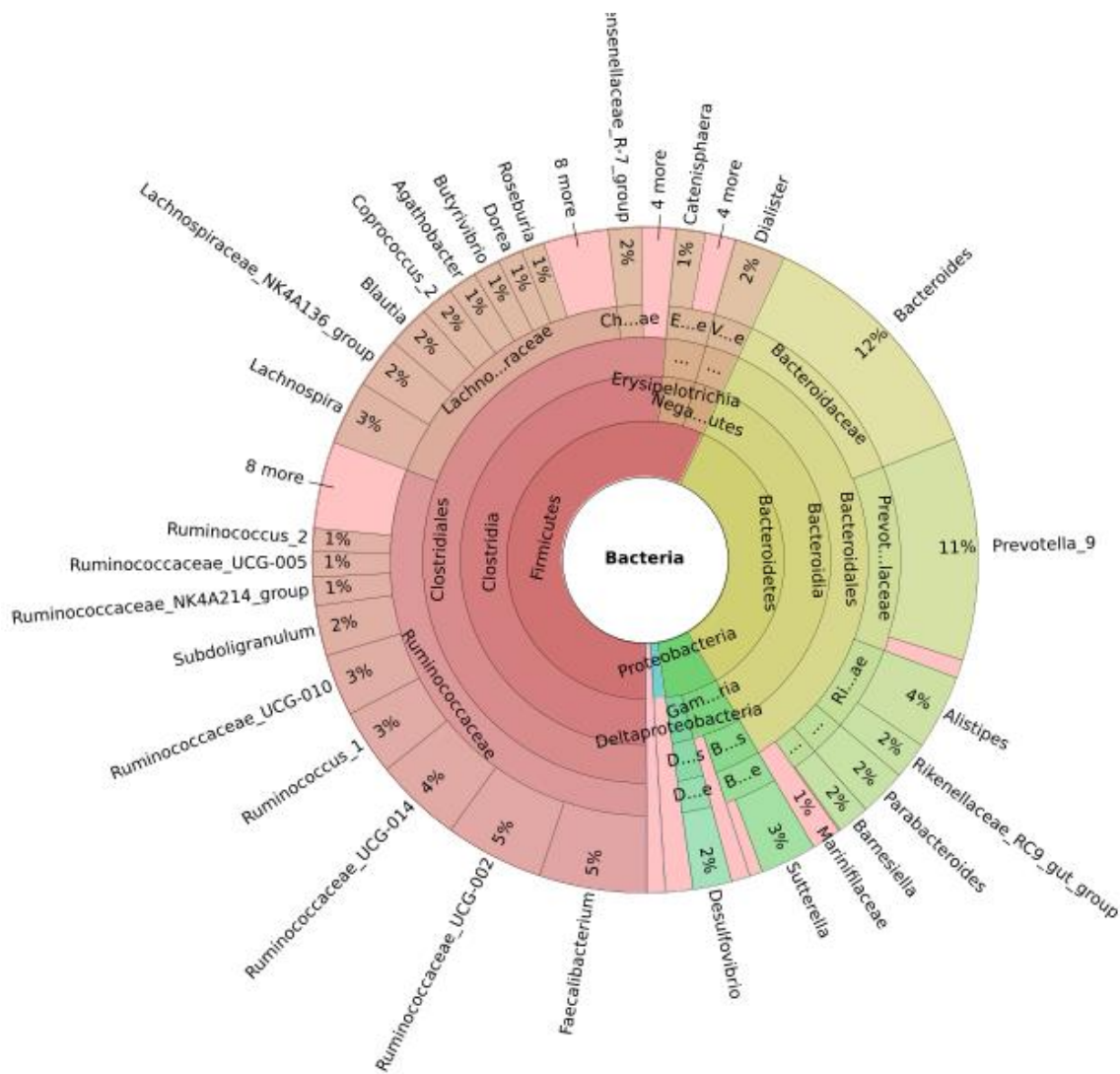


Fig 3.2 DADA2 – taxon relative abundance (Genus level)

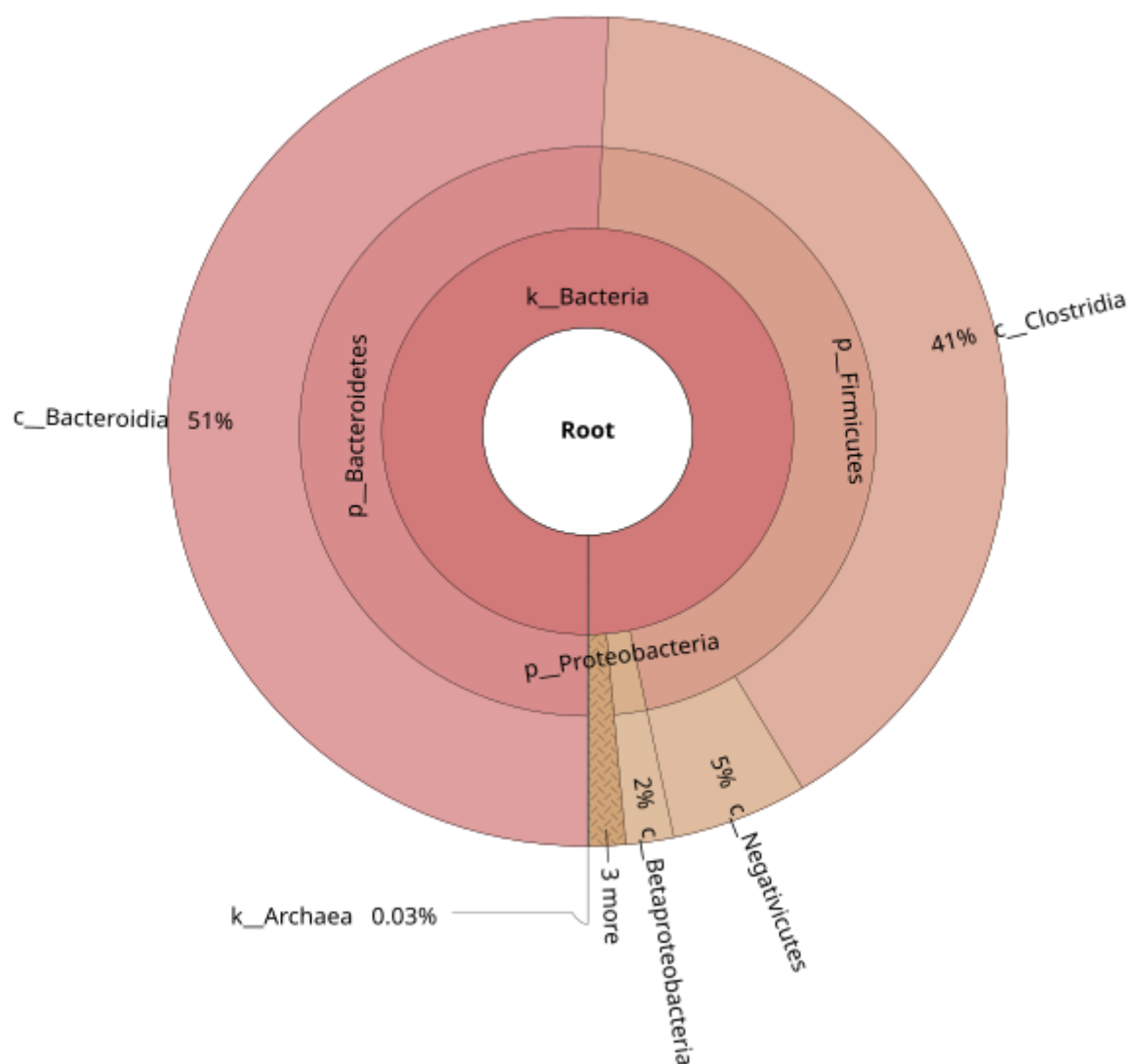


Fig 3.3 Kraken 2 – taxon relative abundance (Class level)

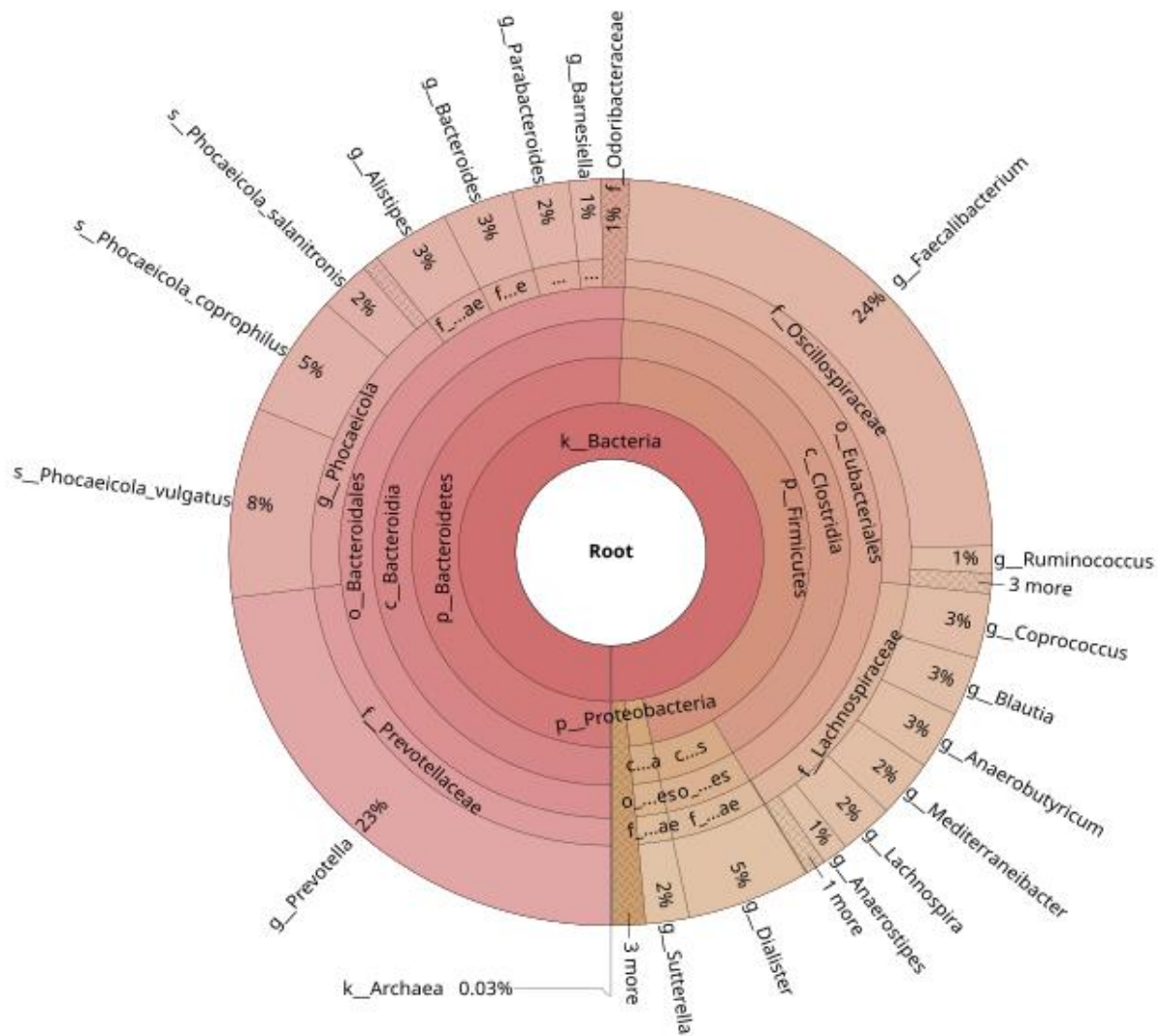


Fig 3.4 Kraken – taxon relative abundance (Genus level)

In the high level Krona views, both pipelines identify Firmicutes/Clostridia and Bacteroidetes/Bacteroidia as the dominant groups in SRR13569534. In DADA2 (plot 1) the largest fractions are Clostridia 51% and Bacteroidia 35%, with smaller contributions from Gammaproteobacteria (4%), Deltaproteobacteria (2%), Negativicutes (2%) and Erysipelotrichia (3%). Kraken2 with Bracken (plot 3) shows the same main players but with shifted proportions: Bacteroidia 51% and Clostridia 41%, plus Negativicutes 5% and a small Betaproteobacteria 2% (and a tiny Archaea signal). The main difference in “level of resolution” is that DADA2 typically stabilizes results at genus level for 16S ASVs (plot 2), while Kraken2 often pushes deeper toward species because it assigns reads by k-mers against genome databases (plot 4). That’s why many species names in Kraken2 could appear (e.g. multiple *Phocaeicola* species), whereas DADA2 shows more genus-level labels (e.g., *Prevotella*_9, *Bacteroides*, *Alistipes*, groups within *Lachnospiraceae*/*Ruminococcaceae*).

Taxon abundancies differ between the methods. One of the example is the Bacteroidia/Clostridia balance: DADA2 suggests more Clostridia (51%) and less Bacteroidia (35%), while Kraken2 suggests more Bacteroidia (51%) and less Clostridia (41%). At finer resolution, DADA2 shows *Prevotella*_9 11% and *Bacteroides* ~2% among the top genera (plot 2), while Kraken2 highlights *Prevotella* 23% and *Faecalibacterium* 24% as major groups (plot 4), with several *Phocaeicola* species contributing noticeable fractions (e.g., 8%, 5%). These differences are normal and can be explained by different algorithms (DADA2 infers exact ASVs by modeling sequencing errors; Kraken2 classifies reads by k-mer matches), different reference databases and naming conventions (16S reference taxonomy vs genome-based RefSeq taxonomy; *Bacteroides* vs *Phocaeicola* splits are a common example), and different sensitivity to ambiguity (Kraken2 can assign to species when k-mers match well, while DADA2 often remains at genus/family for short variable regions if species are not distinguishable).

6. Conclusion

In sample SRR13569534 (16S rRNA V3–V4, paired-end Illumina), the community was dominated by two major groups: Firmicutes, mainly the class Clostridia, and Bacteroidetes, mainly the class Bacteroidia. In the DADA2 profile, Clostridia accounted for about 51% and Bacteroidia for about 35%, while the Kraken2 with Bracken profile showed the same two dominant groups but with shifted proportions, with Bacteroidia around 51% and Clostridia around 41%.

Overall, both DADA2 and Kraken2 gave a consistent high-level picture of the microbiome, agreeing on the dominant phyla/classes and the presence of smaller contributions from Proteobacteria and other minor classes. The main differences appeared at finer resolution: DADA2 results naturally stabilized at genus-level labels typical for 16S ASV workflows, whereas Kraken2 (especially with Bracken) tended to report more species-level names, which can change how the “top taxa” look even when the broader structure is similar.

The most important decision in the analysis was the trimming and filtering strategy, because it controlled the trade-off between keeping enough reads and removing low-quality tails that would inflate errors. Based on the quality profiles, truncation to 280 bp (forward) and 220 bp (reverse) with expected-error filtering (maxEE 2/3) retained 34,553 of 67,311 read pairs (51%), which removed the problematic low-quality regions while still preserving overlap for successful merging and clean downstream denoising.

For 16S data, DADA2 produced the more readable result, because genus-level ASV-based profiles are usually more directly interpretable for V3–V4 amplicons than long lists of species labels that can depend strongly on the genome database and k-mer matching.